

Structured ASIC Placer Report

1 Introduction

This report summarizes the current implementation and experimental findings for the structured ASIC placer. Two initial placement case studies were evaluated:

- **Rationale-based initial placement:** cells are placed using connectivity-driven estimates derived from fixed pins and already placed cells, followed by placement on the nearest legal site of matching type.
- **Random initial placement:** cells are placed randomly on unused legal sites of matching type.

Both case studies were followed by simulated annealing (SA) using legal typed moves:

- swap between two cells of the same type
- move of one cell to an empty legal site of the same type

The objective metric throughout the experiments is total wire length, measured as half-perimeter wire length (HPWL / TWL).

2 Experimental Setup

The SA configuration used in the current experiments is:

- Initial temperature: $500 \times \text{initial cost}$
- Final temperature: $5 \times 10^{-5} \times \frac{\text{initial cost}}{\text{number of nets}}$
- Moves per temperature: $20 \times \text{number of movable cells}$
- Cooling rates evaluated: 0.75, 0.80, 0.85, 0.90, and 0.95

The benchmark suite contains five provided designs:

- `design_1_small`
- `design_2_medium`
- `design_3_large`
- `design_4_dense`
- `design_5_extreme`

3 Implementation Summary

3.1 Rationale-Based Initial Placement

The rationale-based initial placer starts with only fixed pins having known coordinates. It repeatedly selects an unplaced cell that has known connected anchors, estimates the cell position using the average of connected pin and already placed cell coordinates, and places the cell on the nearest unused legal site of the same type. After placement, the chosen site location becomes the cell's known position for later estimates. If a disconnected region is encountered, the algorithm falls back to a legal placement from a neutral estimate so that the process can continue.

3.2 Random Initial Placement

The random initial placer ignores connectivity. It groups legal sites by type, shuffles them, and assigns each movable cell to a random unused legal site of matching type. This provides a valid but much weaker starting point for SA.

3.3 Simulated Annealing

The SA stage starts from a legal typed placement and applies two legal move types:

- same-type swap between two cells
- same-type move from a cell to an empty legal site

The cost function is total HPWL. Cost updates are localized by recomputing only nets incident to the moved cells, rather than recomputing the full design cost after every attempted move.

4 Temperature vs. TWL at Cooling Rate 0.95

The first experiment tracks TWL as temperature decreases for both initial placement strategies using cooling rate 0.95.

4.1 Key Findings

- The rationale-based initializer always starts from a much better initial TWL than the random initializer.
- The random initializer shows much larger percentage improvement because it starts from a substantially worse placement.
- After SA, the final TWL values of the two case studies become much closer, which indicates that SA is doing substantial optimization work.

4.2 Numerical Summary

Design	Init. Rat.	Final Rat.	Init. Rand.	Final Rand.
design_1_small	2706	2208	6747	2196
design_2_medium	4293	3449	13192	3463
design_3_large	8051	6133	24057	6189
design_4_dense	14809	10899	40018	10905
design_5_extreme	24781	17980	57950	17838

4.3 Discussion

The rationale-based initial placement provides a stronger starting solution on every benchmark because it uses connectivity information before SA begins. In contrast, the random initial placement begins with very poor TWL, especially on the larger and denser benchmarks. However, SA is still able to reduce the random baseline dramatically and bring it close to the rationale-based final result.

This behavior shows two things:

- a better initializer improves the starting quality immediately
- SA is strong enough to recover a large amount of quality even from a weak random baseline

5 Cooling Rate vs. TWL

The second experiment compares the final TWL obtained under the required cooling rates:

0.75, 0.80, 0.85, 0.90, 0.95

5.1 Key Findings

- Across all five designs and both initial placement modes, the best final TWL was obtained at cooling rate 0.95.
- Lower cooling rates finished in fewer temperature steps, which reduces runtime.
- The faster schedules produced worse final TWL because the search cooled too aggressively.

5.2 Best Cooling Rate per Design

Design	Best CR (Rationale)	Best CR (Random)
design_1_small	0.95	0.95
design_2_medium	0.95	0.95
design_3_large	0.95	0.95
design_4_dense	0.95	0.95
design_5_extreme	0.95	0.95

5.3 Discussion

The cooling-rate sweep shows the expected quality-runtime tradeoff. A lower cooling rate reduces the number of temperature levels, so the algorithm finishes sooner. However, this also reduces the amount of search and exploration, which hurts final placement quality. In the current implementation, cooling rate 0.95 is the strongest quality setting among the required values and should be used when final TWL matters more than runtime.

6 Runtime Efficiency Discussion

The current code uses several data-structure choices that make the SA implementation practical:

- A reverse net-membership map stores, for each component, the nets incident to it.

- A current position map stores the location of every placed component.
- Legal sites are grouped by type, which makes move generation and legality checks simple.
- Only affected nets are re-evaluated after a move, instead of recomputing total HPWL for the entire design.

These choices significantly reduce the amount of work per move. The experiments also show that runtime is dominated by the number of temperature steps and attempted moves. This is why higher cooling rates achieve better quality but require more computation.

7 Overall Conclusions

Based on the current results, the following conclusions can be drawn:

- The rationale-based initial placement is clearly better than random placement as a starting point.
- Random placement is still useful as a baseline because it highlights how much SA improves the solution.
- SA is effective in both case studies and is able to reduce TWL substantially.
- In the current implementation, cooling rate 0.95 consistently produces the best final TWL across all five benchmarks.

8 Graphs

The figures below are included from the experiment SVG graph files. The SVG files were converted to PDF format for reliable LaTeX compilation.

8.1 Design 1: Small

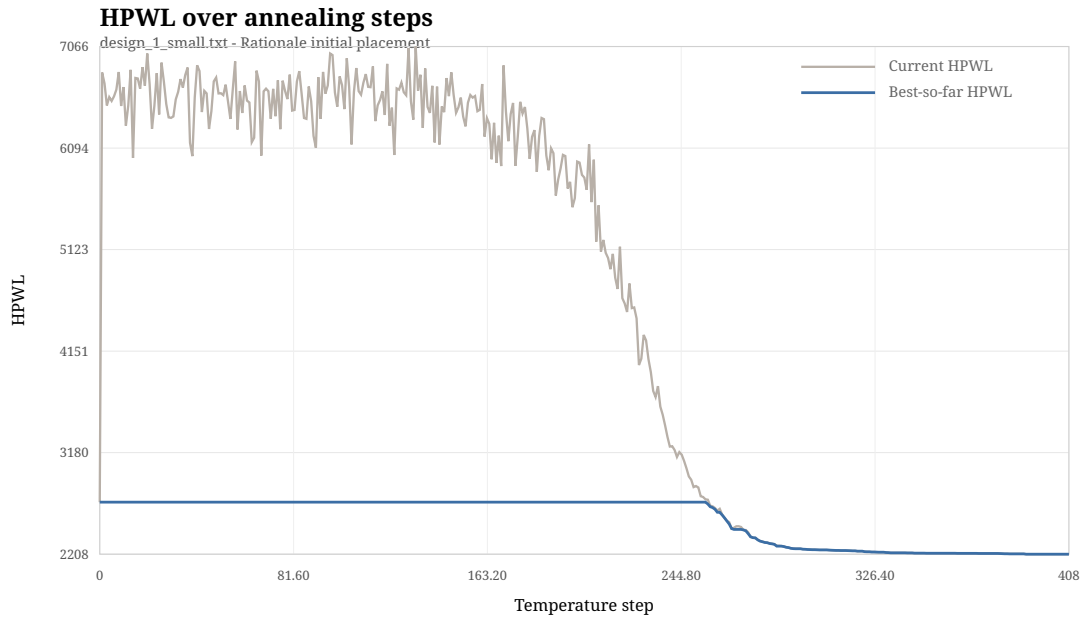


Figure 1: Temperature vs. TWL for design_1_small using rationale-based initial placement.

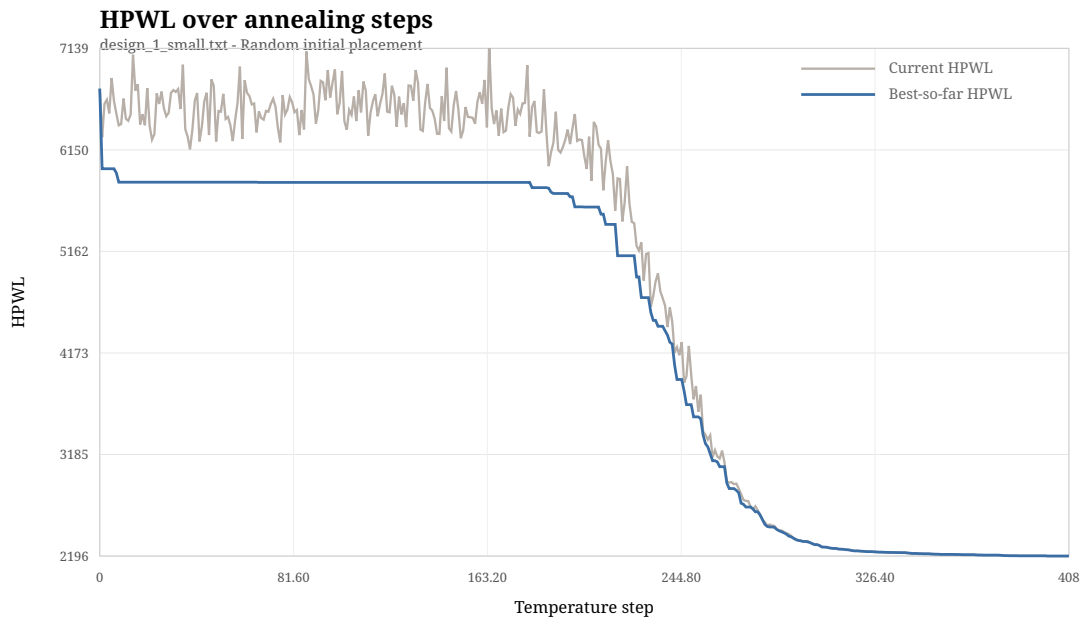


Figure 2: Temperature vs. TWL for design_1_small using random initial placement.

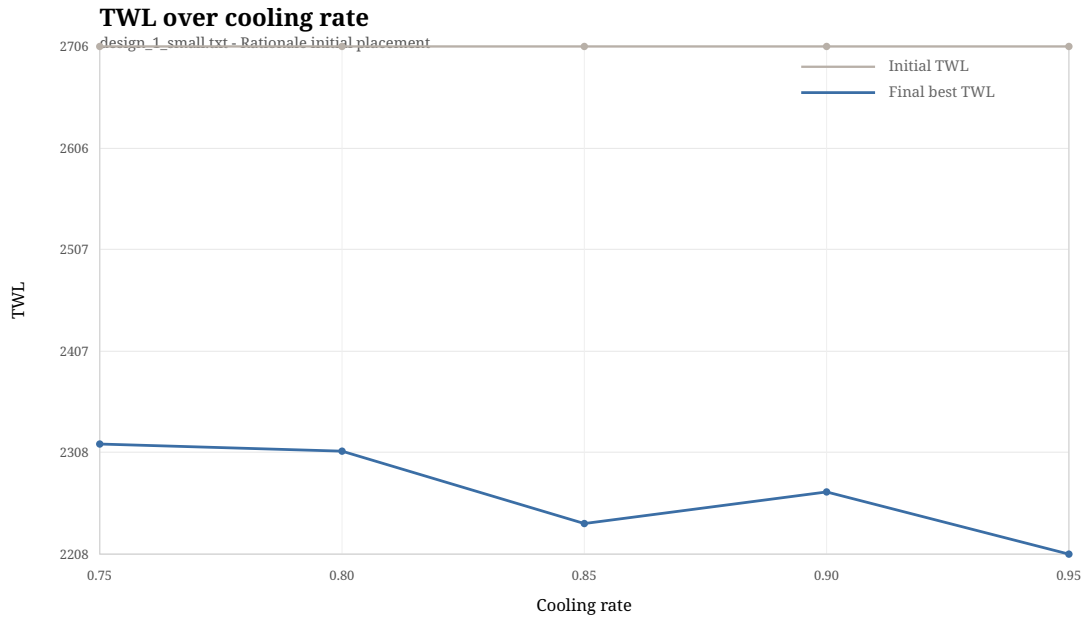


Figure 3: Cooling Rate vs. TWL for design_1_small using rationale-based initial placement.

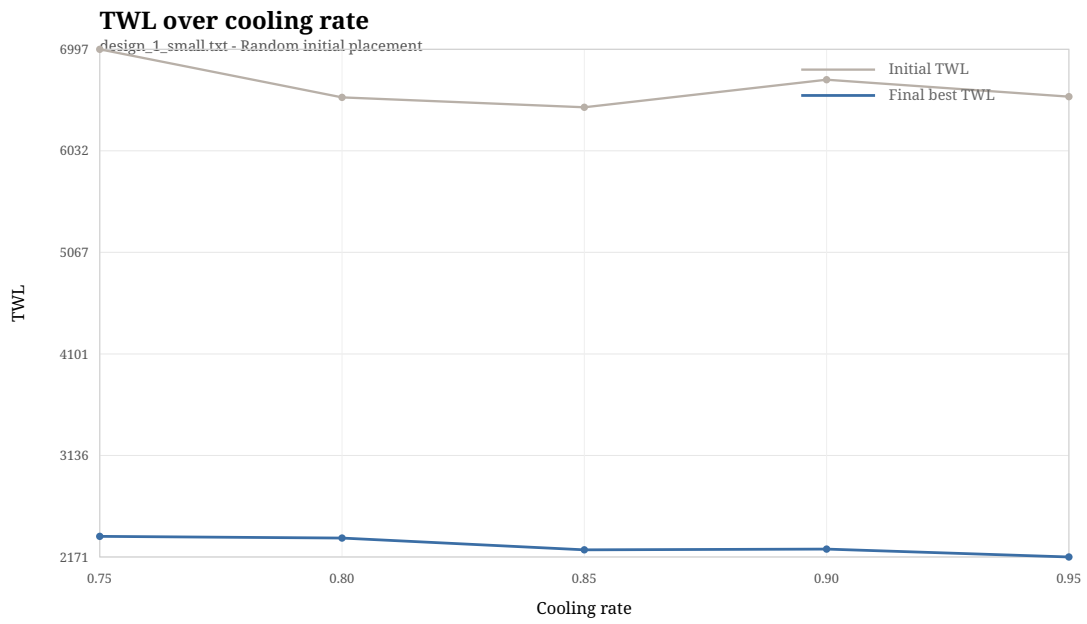


Figure 4: Cooling Rate vs. TWL for design_1_small using random initial placement.

8.2 Design 2: Medium

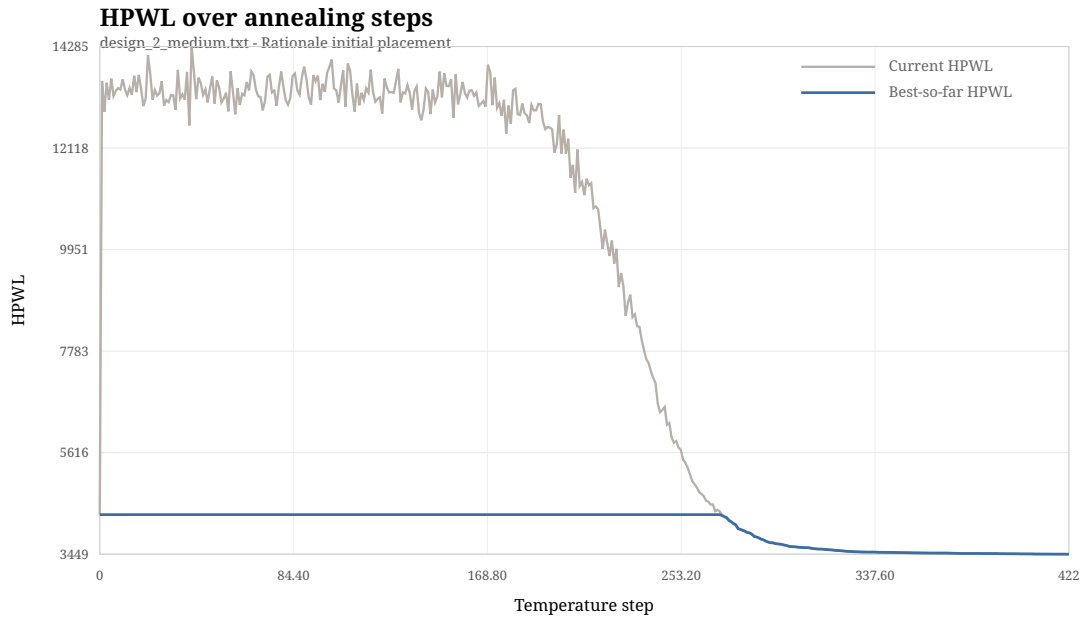


Figure 5: Temperature vs. TWL for design_2_medium using rationale-based initial placement.

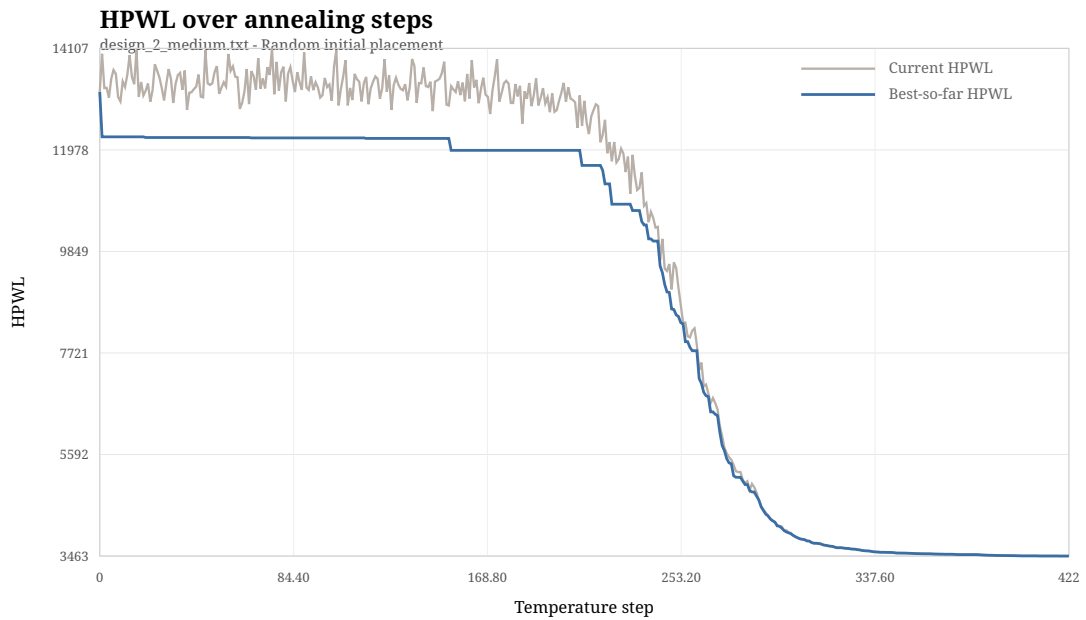


Figure 6: Temperature vs. TWL for design_2_medium using random initial placement.

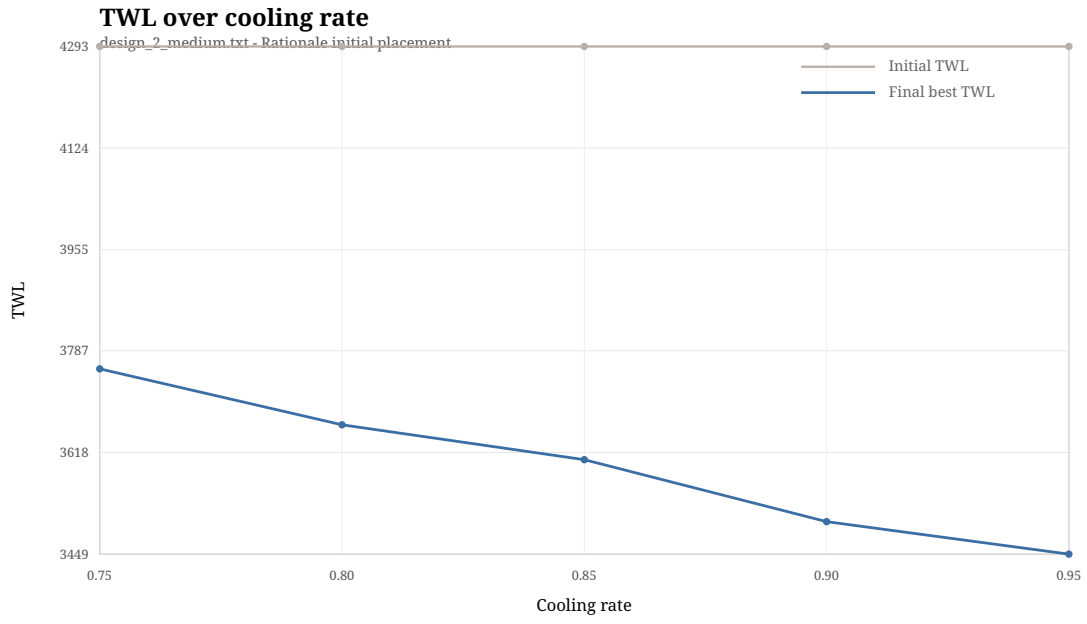


Figure 7: Cooling Rate vs. TWL for design_2_medium using rationale-based initial placement.

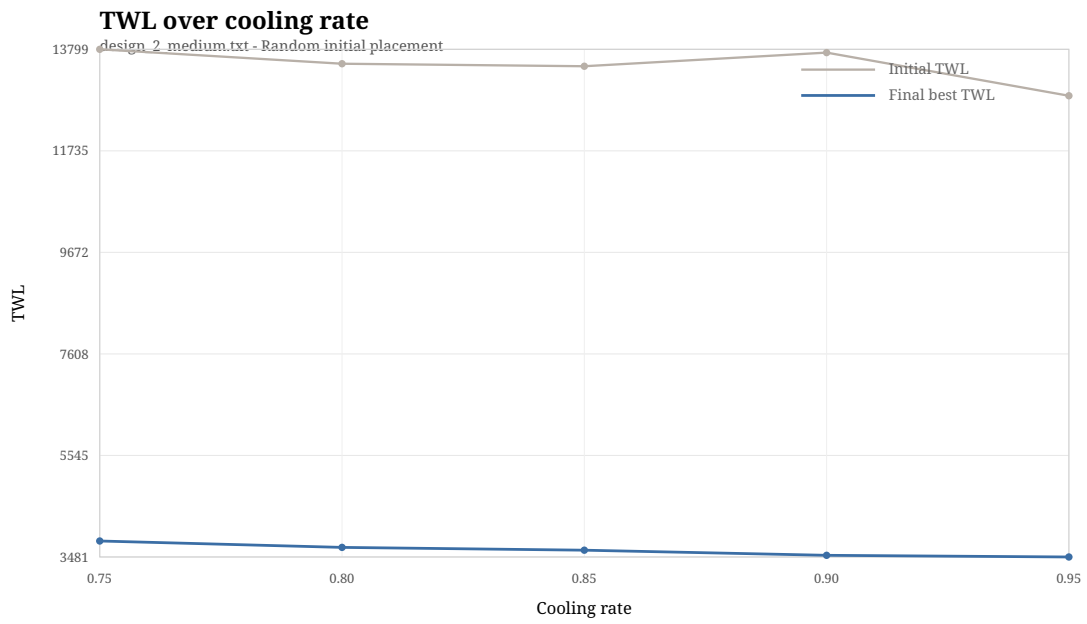


Figure 8: Cooling Rate vs. TWL for design_2_medium using random initial placement.

8.3 Design 3: Large

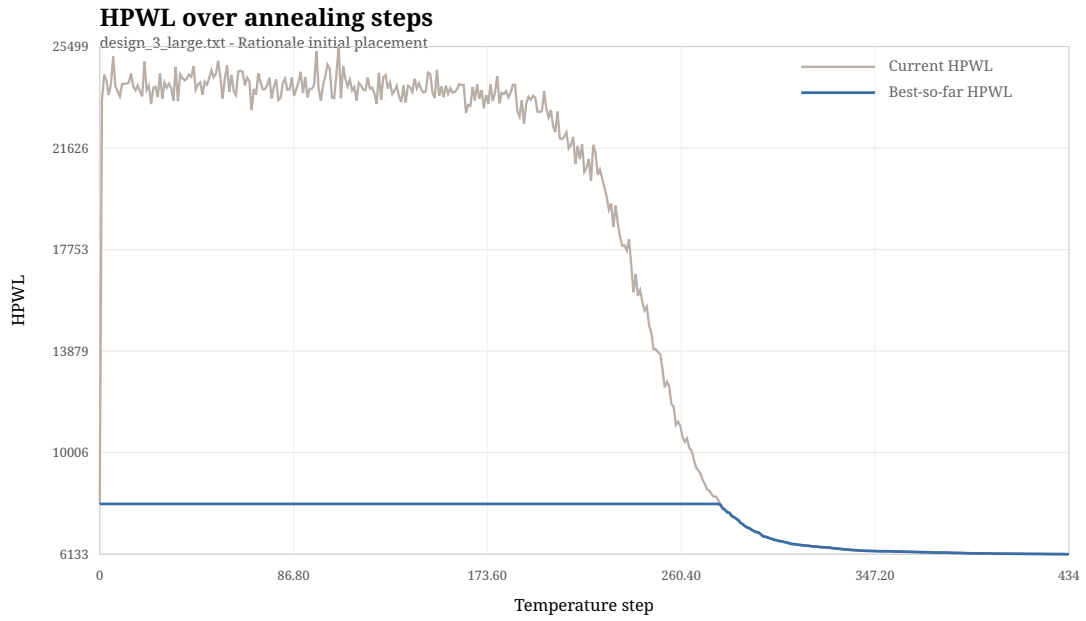


Figure 9: Temperature vs. TWL for design_3_large using rationale-based initial placement.

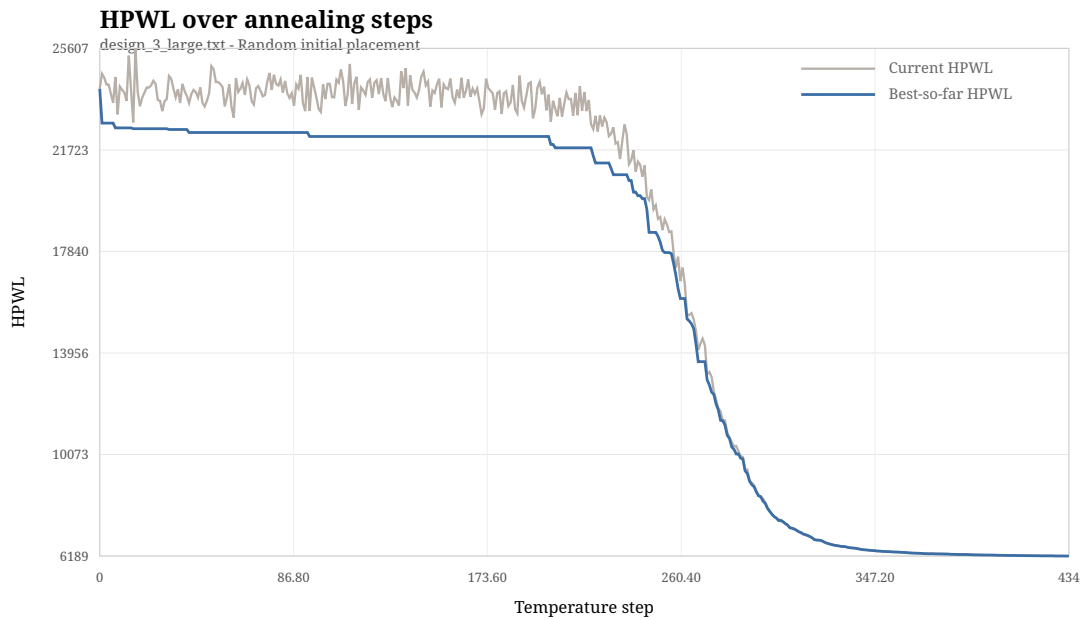


Figure 10: Temperature vs. TWL for design_3_large using random initial placement.

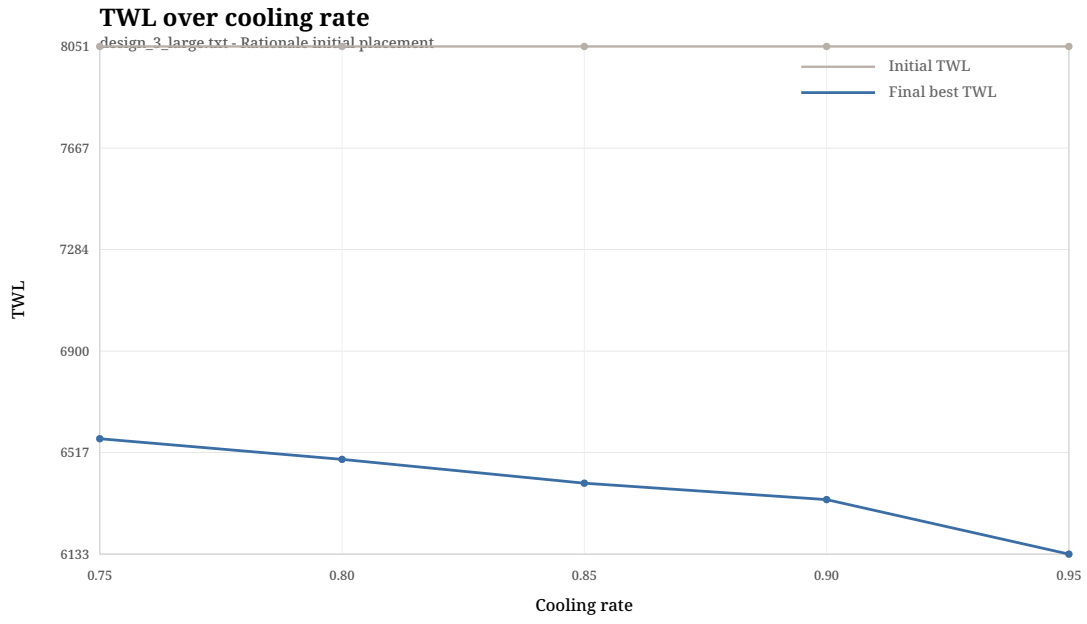


Figure 11: Cooling Rate vs. TWL for design_3_large using rationale-based initial placement.

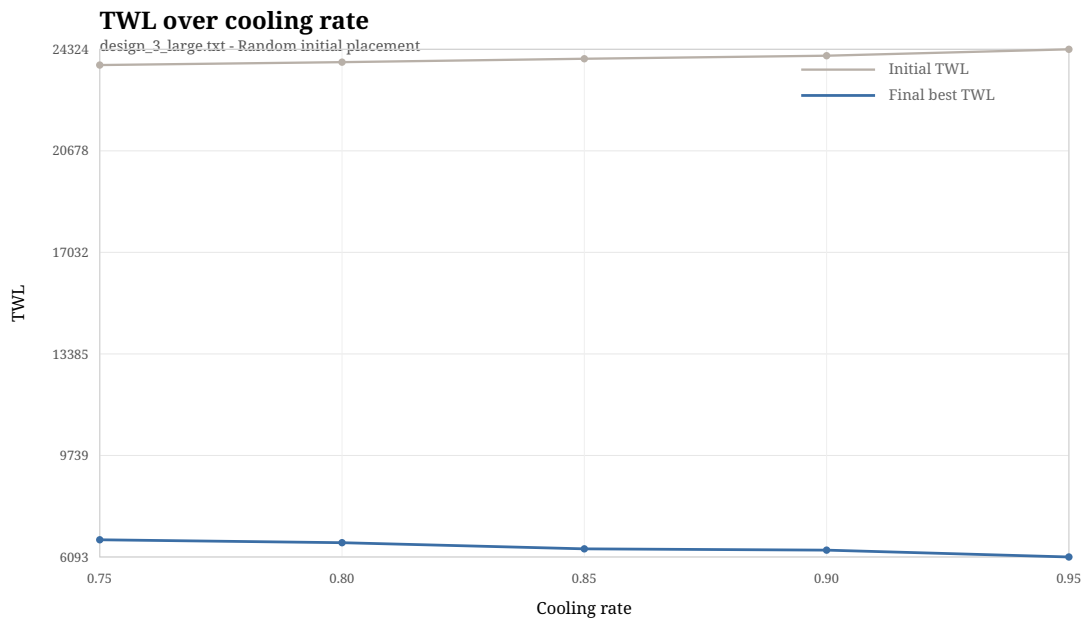


Figure 12: Cooling Rate vs. TWL for design_3_large using random initial placement.

8.4 Design 4: Dense

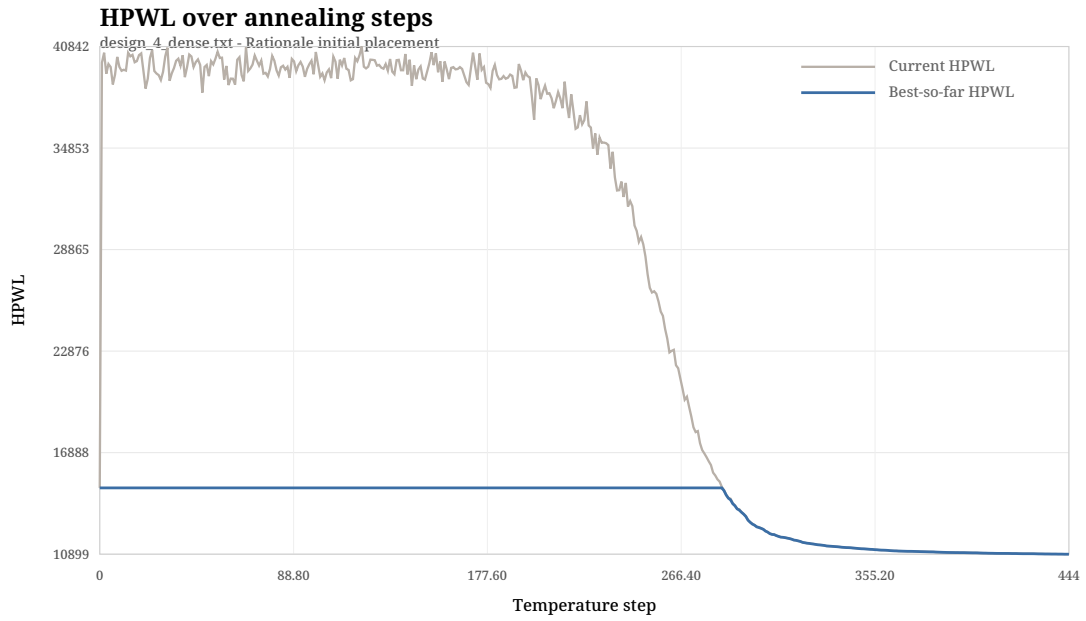


Figure 13: Temperature vs. TWL for design_4_dense using rationale-based initial placement.

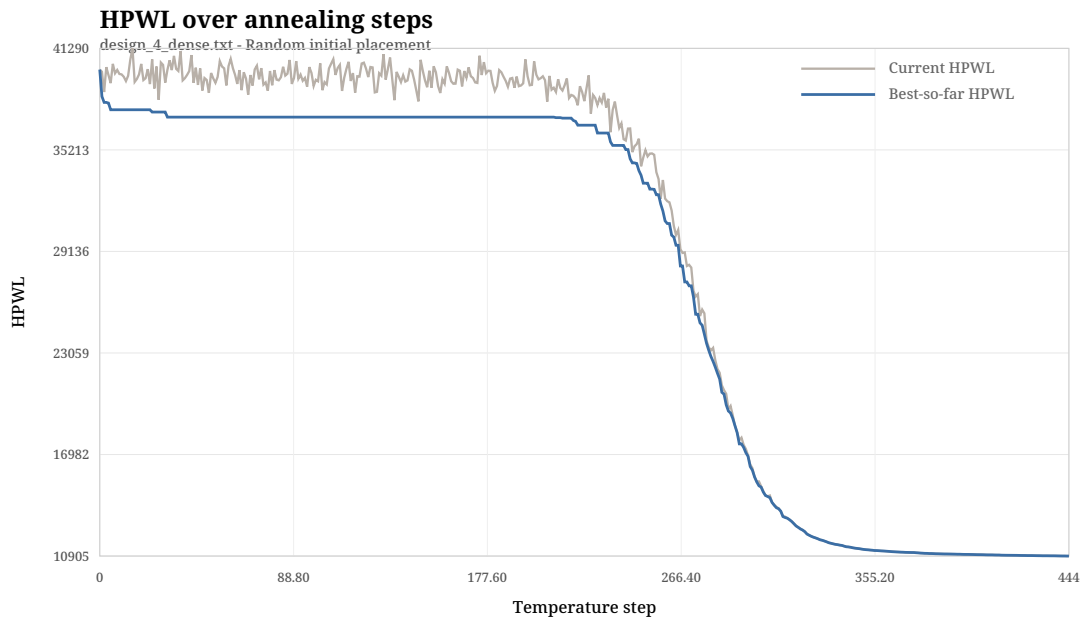


Figure 14: Temperature vs. TWL for design_4_dense using random initial placement.

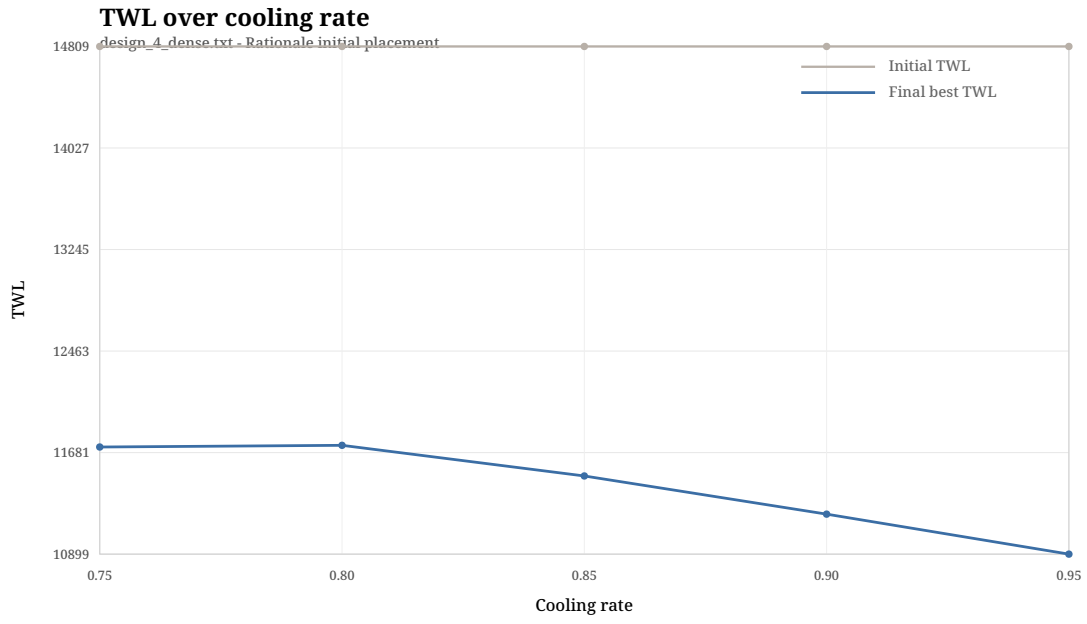


Figure 15: Cooling Rate vs. TWL for design_4_dense using rationale-based initial placement.

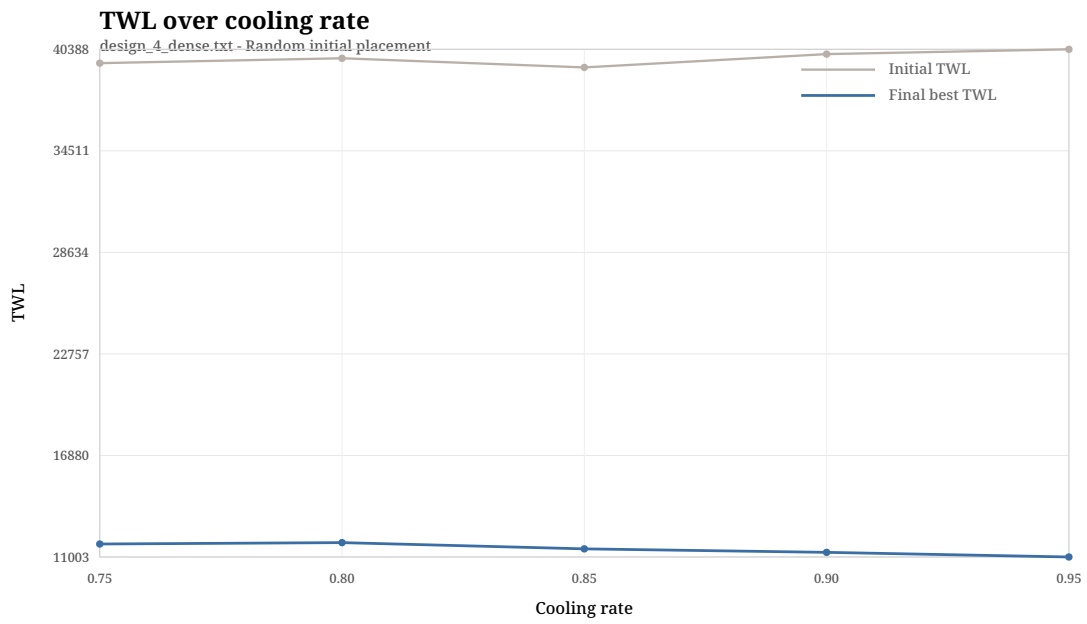


Figure 16: Cooling Rate vs. TWL for design_4_dense using random initial placement.

8.5 Design 5: Extreme

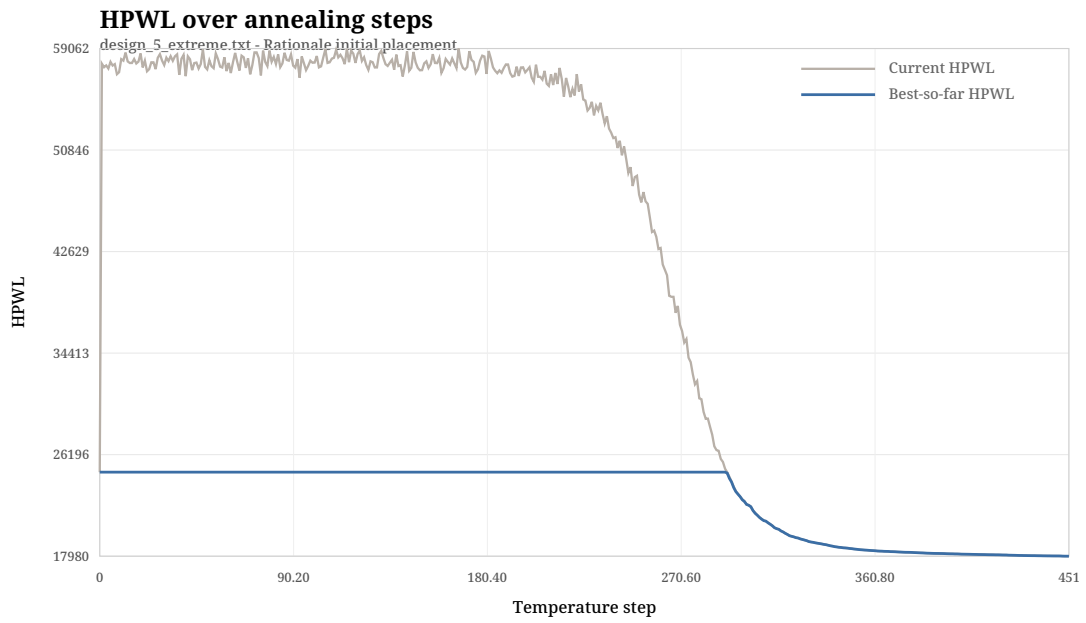


Figure 17: Temperature vs. TWL for design_5_extreme using rationale-based initial placement.

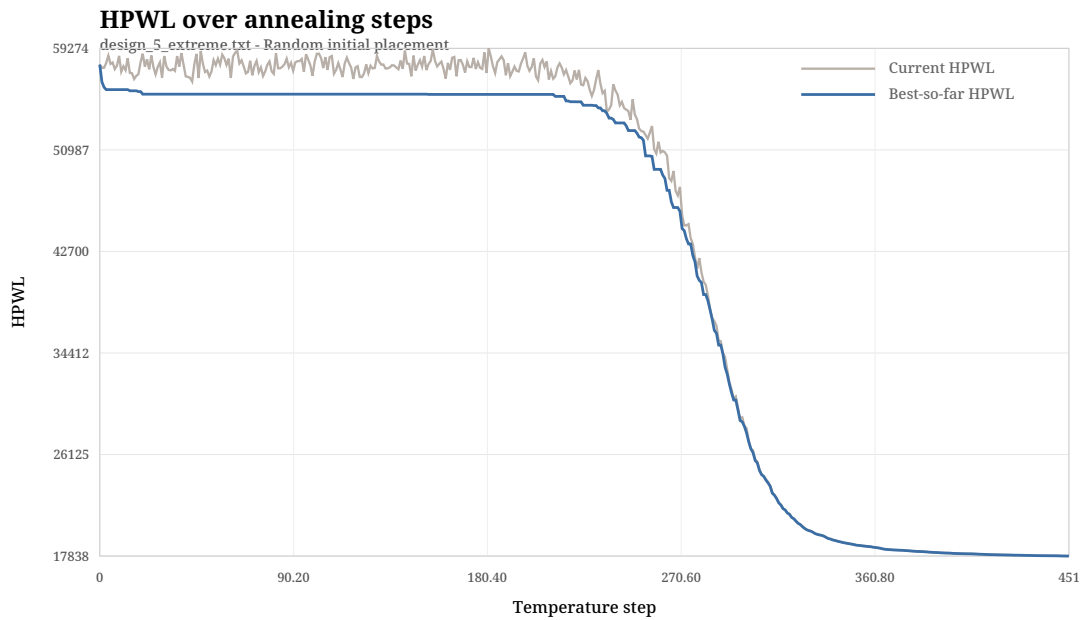


Figure 18: Temperature vs. TWL for design_5_extreme using random initial placement.

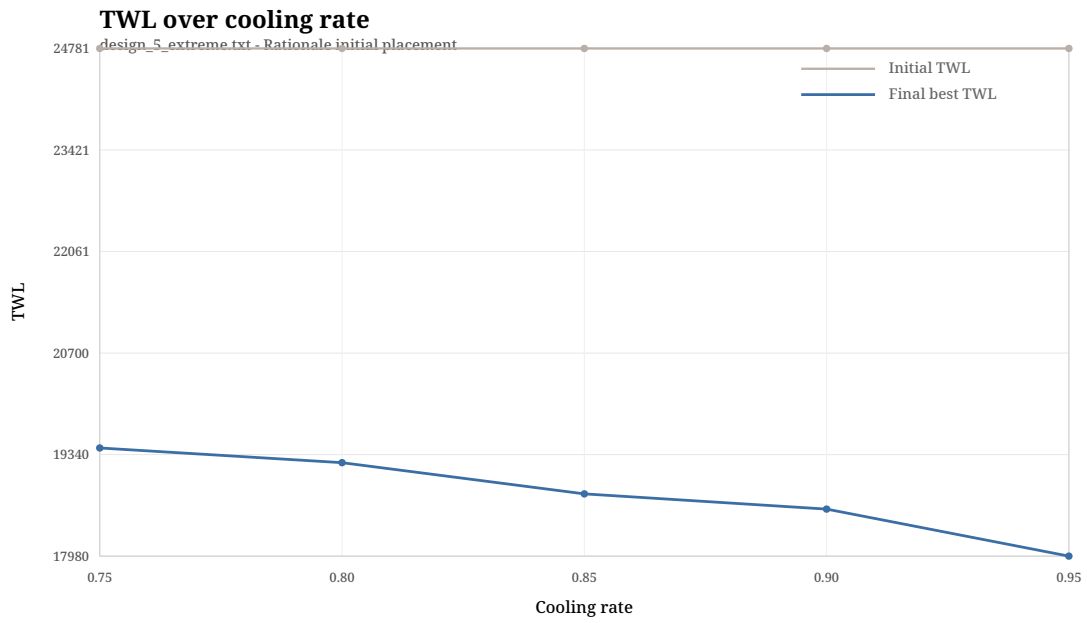


Figure 19: Cooling Rate vs. TWL for design_5_extreme using rationale-based initial placement.

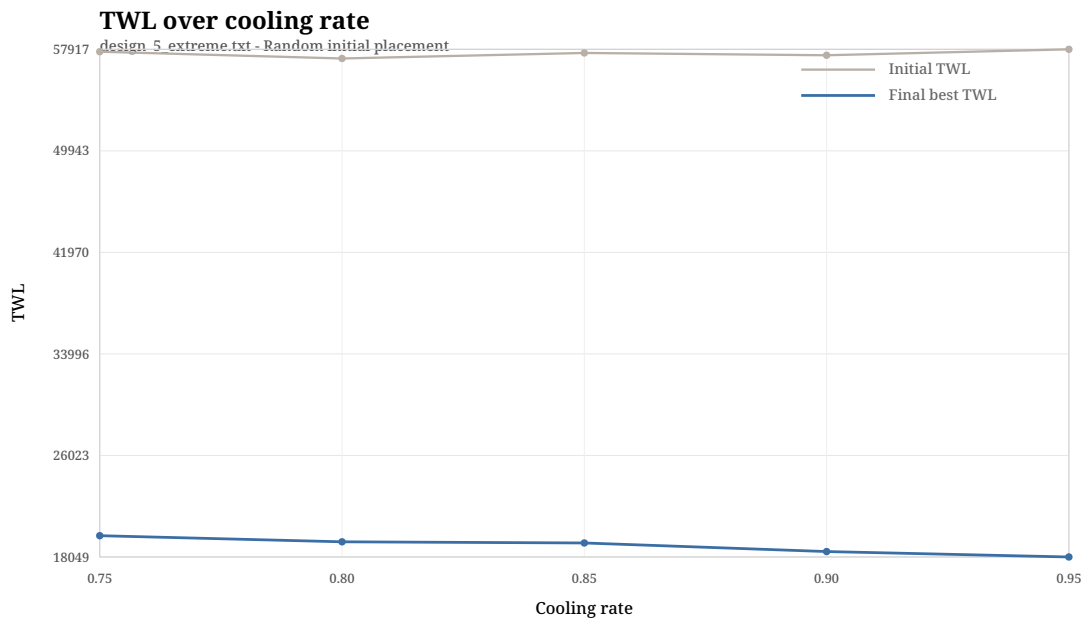


Figure 20: Cooling Rate vs. TWL for design_5_extreme using random initial placement.